

Script to Present in Class

Topic: AI Doctor – Intelligent Disease Prediction and Triage System

Slide 1 — Title Slide: AI Doctor

Say:

Good afternoon everyone.

My name is **Jemi Jose**, and today I will be presenting my project called **AI Doctor**, which is an intelligent disease prediction and triage system.

This project demonstrates how machine learning and web technologies can work together to help users understand their symptoms and get early health insights.

Slide 2 — The Healthcare Waiting Journey

Say:

In healthcare, many people do not have enough medical knowledge to understand their symptoms correctly.

They often face long waiting times and limited access to doctors.

So, this project aims to provide a faster and more efficient way for users to input symptoms and receive preliminary health guidance.

The goal is to reduce delays and improve early decision-making.

Slide 3 — Automated Preliminary Analysis at Scale

Say:

AI Doctor is a **full-stack web application** powered by machine learning.

It allows users to enter their symptoms through an easy-to-use chat interface.

Then, the system analyzes the symptoms and predicts the most probable disease.

It also provides a **confidence score**, which shows how certain the prediction is.

This helps users get quick health awareness.

Slide 4 — Evaluating Assessment Methodologies

Say:

Here, we compare traditional health assessment methods with the AI Doctor system.

Traditional methods can be slow and limited by location and availability.
But AI Doctor provides:

- Real-time predictions
- 24/7 accessibility
- Better handling of complex symptom data
- More consistent results

This makes the system faster and more accessible for users.

Slide 5 — The Assistive Triage Layer

Say:

It is important to understand that AI Doctor is **not a replacement for doctors**.

Instead, it acts as an assistive tool that provides early health insights.

The system helps users decide whether they need professional medical consultation.

So, it supports doctors rather than replacing them.

Slide 6 — The 3-Tier API Request Cycle

Say:

This slide shows the system architecture.

The application uses a **three-tier design**, which separates:

- The user interface
- The backend server
- The machine learning model

When a user enters symptoms, the system sends a request to the server, processes the data, and returns the prediction.

This design makes the system scalable and efficient.

Slide 7 — Diagnostic Breakdown of the Tech Stack

Say:

This project uses different technologies for different parts of the system.

For the **frontend**, we use:

React.js and CSS to create the user interface.

For the **backend**, we use:

Python and Flask to manage communication between the user and the model.

For **machine learning**, we use:

Scikit-learn and a Random Forest algorithm to analyze symptom data and make predictions.

Slide 8 — The Machine Learning Data Refinery

Say:

This slide shows the steps used to build the machine learning model.

The process includes:

1. Collecting data from a Kaggle dataset
2. Cleaning and preprocessing the data
3. Training the model
4. Evaluating its accuracy
5. Integrating the model into the web application

These steps ensure that the model produces reliable predictions.

Slide 9 — From Symptom to Prediction

Say:

The system uses a **Random Forest algorithm**, which is very effective for handling complex data.

Each symptom is converted into a binary value, such as:

0 for absence

1 for presence

The algorithm then analyzes these values to identify patterns and predict the most likely disease.

This process allows the system to make accurate predictions quickly.

Slide 10 — High-Accuracy Real-Time Inference

Say:

This slide shows an example of the system in action.

When a user enters symptoms like fever and cough, the system analyzes the data and predicts a disease.

For example:

Predicted disease: Common Cold

Confidence score: 89 percent

The prediction happens instantly, which makes the system fast and user-friendly.

Slide 11 — System Boundaries and Clinical Realities

Say:

Although the system is powerful, it also has limitations.

For example:

- It cannot replace professional medical diagnosis
- It may struggle with complex or unclear symptoms
- It depends on the quality of the training data

So, the system should be used only as an early guidance tool.

Slide 12 — Future Evolution of the Platform

Say:

In the future, this system can be improved in several ways.

For example:

- Adding Natural Language Processing to understand complex sentences
- Training the model on more diverse datasets
- Adding voice input features
- Deploying the system on cloud platforms for global access

These improvements can make the system more intelligent and accessible.

Slide 13 — Conclusion: Bridging the Healthcare Information Gap

Say:

In conclusion, AI Doctor shows how artificial intelligence can improve healthcare accessibility.

By combining machine learning and web technologies, the system provides fast and reliable early health insights.

This project demonstrates the growing role of AI in supporting healthcare systems and helping people make informed decisions.

Thank you for listening.

I am happy to answer any questions.